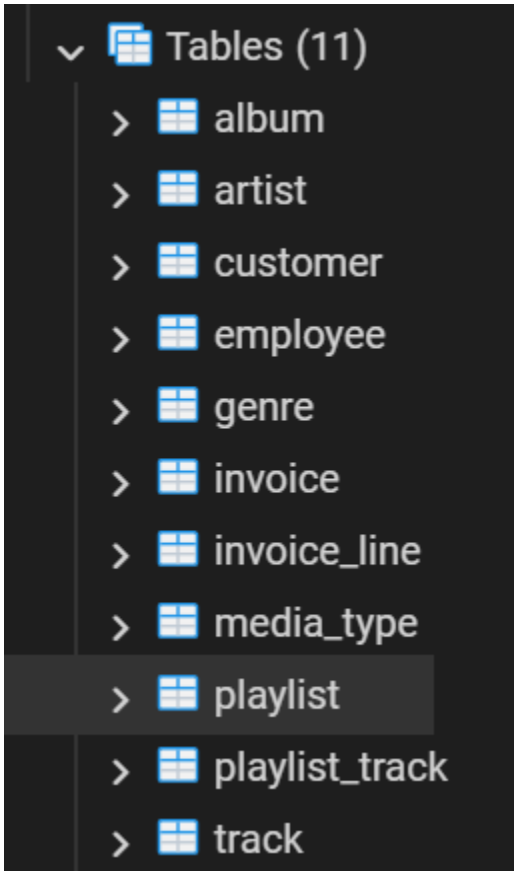
Music Store Analysis Using PostgreSql

This project analyzes a relational music store database using PostgreSQL. The goal is to extract business insights related to customer behavior, sales performance, and music preferences using SQL queries.

Dataset Overview

The dataset contains multiple related tables including:



These tables help analyze **music sales, customer purchases, and artist popularity**.

Business Questions Solved

1. Who is the **senior most employee** based on job title?
2. Which **countries have the most invoices**?
3. What are the **top 3 invoice totals**?
4. Which **city generated the highest revenue**?
5. Who is the **best customer (highest spending)**?
6. Identify **Rock music listeners**.
7. Find **top 10 artists with the most rock tracks**.
8. Identify **songs longer than the average song length**.
9. Calculate **amount spent by each customer on artists**.
10. Find the **most popular genre in each country**.
11. Identify the **top customer by spending in each country.**

Query Analysis

- Senior Most Employee

Query Query History

```sql
SELECT title, last_name, first_name
FROM employee
ORDER BY levels DESC
LIMIT 1
```

Data Output Messages Notifications

Showing rows: 1 to 1

	title character varying (50)	last_name character (50)	first_name character (50)
1	Senior General Manag...	Madan	Mohan

- This query identifies the employee with the **highest job level in the company hierarchy**.

- Country with Most Invoices

Query Query History

```sql
SELECT COUNT(*) AS c, billing_country
FROM invoice
GROUP BY billing_country
ORDER BY c DESC
```

Data Output Messages Notifications

Showing rows: 1 to 24

	c bigint	billing_country character varying (30)
1	131	USA
2	76	Canada
3	61	Brazil
4	50	France
5	41	Germany
6	30	Czech Republic
7	29	Portugal
8	28	United Kingdom
9	21	India
10	13	Chile

Total rows: 24 Query complete 00:00:00.151

- The analysis shows that the **USA generates the highest number of invoices (131)**, making it the largest market for the music store. Canada and Brazil also contribute significantly to the total number of transactions.

- City with Highest Revenue

Query Query History

```sql
SELECT billing_city,SUM(total) AS InvoiceTotal
FROM invoice
GROUP BY billing_city
ORDER BY InvoiceTotal DESC
LIMIT 1;
```

Data Output Messages Notifications

Showing rows: 1 to 1

	billing_city character varying (30)	invoicetotal double precision
1	Prague	273.24000000000007

- The analysis shows that **Prague generated the highest total revenue (273.24)** from music purchases in the dataset.
- Based on the revenue analysis, the company could prioritize **marketing campaigns or live music events in Prague** to further increase sales and customer engagement.

- # Best Customer

Query Query History

```sql
SELECT customer.customer_id, first_name, last_name, SUM(total) AS total_spending
FROM customer
JOIN invoice ON customer.customer_id = invoice.customer_id
GROUP BY customer.customer_id
ORDER BY total_spending DESC
LIMIT 1;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1

	customer_id [PK] integer	first_name character (50)	last_name character (50)	total_spending double precision
1	5	R	Madhav	144.54000000000002

- The analysis shows that **Customer ID 5 (Madhav R)** is the **highest spending customer**, with a total purchase amount of **144.54**.
- This indicates that this customer contributes significantly to the store's revenue and could be considered a **high-value customer**. Businesses often target such customers with **loyalty programs, exclusive offers, or personalized promotions** to maintain long-term engagement.

- # Popular Genre in Country

Query Query History

```sql
WITH popular_genre AS
(
    SELECT COUNT(invoice_line.quantity) AS purchases, customer.country, genre
    ROW_NUMBER() OVER(PARTITION BY customer.country ORDER BY COUNT(invoice_li
    FROM invoice_line
    JOIN invoice ON invoice.invoice_id = invoice_line.invoice_id
    JOIN customer ON customer.customer_id = invoice.customer_id
    JOIN track ON track.track_id = invoice_line.track_id
    JOIN genre ON genre.genre_id = track.genre_id
```

Data Output Messages Notifications

Showing rows: 1 to 24 Page No

	purchases bigint	country character varying (50)	name character varying (120)	genre_id character varying (50)	rowno bigint
1	17	Argentina	Alternative & Punk	4	1
2	34	Australia	Rock	1	1
3	40	Austria	Rock	1	1
4	26	Belgium	Rock	1	1
5	205	Brazil	Rock	1	1
6	333	Canada	Rock	1	1
7	61	Chile	Rock	1	1

Total rows: 24 Query complete 00:00:00.071

The analysis reveals that **Rock is the most popular music genre in the majority of countries** in the dataset. Countries such as **Canada, Brazil, and Australia show a particularly high number of rock music purchases**, indicating strong demand for this genre.

This insight can help the music store:

- Focus **marketing campaigns around rock music**
- Promote **rock artists and albums**
- Organize **genre-specific promotions or playlists**

Key Insights

1. Strong Market in the USA

The **USA generated the highest number of invoices (131)**, indicating that it is the largest and most active market for the music store.

2. Prague Generates the Highest Revenue

Among all cities, **Prague produced the highest revenue (273.24)** from music purchases, suggesting a strong customer base in this city.

3. High Value Customers Exist

Customer **Madhav R (Customer ID 5)** is the **highest spending customer**, with a total purchase amount of **144.54**, showing that a small group of customers contributes significantly to revenue.

4. Rock Music Dominates Globally

Analysis of genre purchases across countries shows that **Rock is the most popular genre in most countries**, especially in markets like **Canada, Brazil, and Australia**.

5. Genre Preference Varies Across Countries

Although Rock dominates globally, some countries show preference for **alternative genres**, indicating regional music taste differences.

Recommendation

1. Focus Marketing on High Performing Markets

Since the **USA has the highest number of transactions**, marketing campaigns and promotions should prioritize this region.

2. Organize Events in Prague

Because **Prague generates the highest revenue**, it could be an ideal location for **music festivals, concerts, or promotional campaigns**.

3. Retain High Value Customers

Customers who spend large amounts, like **Madhav R**, should be targeted with:

- loyalty programs
- personalized recommendations
- exclusive discounts

to increase retention.

4. Promote Popular Genres

Since **Rock music is the most purchased genre**, the store could:

- highlight rock playlists

- promote rock artists
- recommend rock albums to customers.

5. Localized Genre Promotions

Different countries may prefer different genres, so **region-specific promotions** could increase engagement and sales.

Conclusion

This project demonstrates how **SQL and PostgreSQL can be used to analyze relational databases and extract valuable business insights**. By analyzing customer purchases, invoice data, and music genres, we identified key trends related to **customer behavior, revenue distribution, and music preferences**.

The insights generated from this analysis can help the business make **data-driven decisions related to marketing strategies, customer engagement, and product promotion**.

Overall, this project highlights the importance of **data analysis in understanding business performance and improving decision-making processes**.